

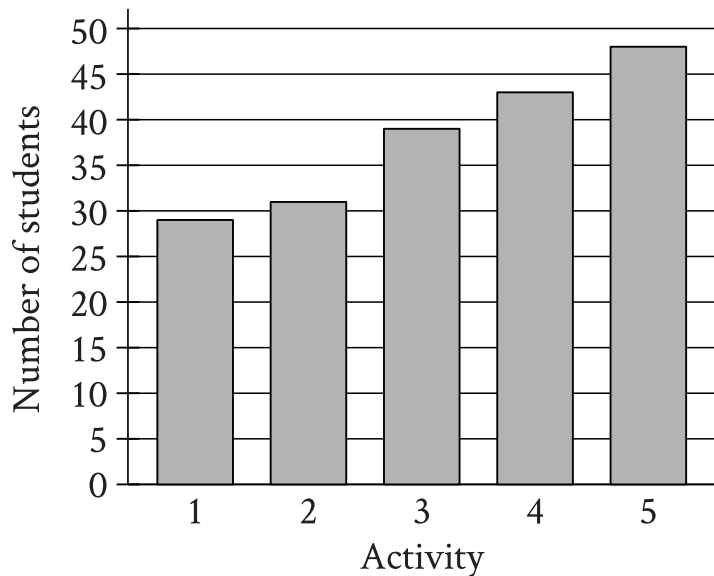
● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

16

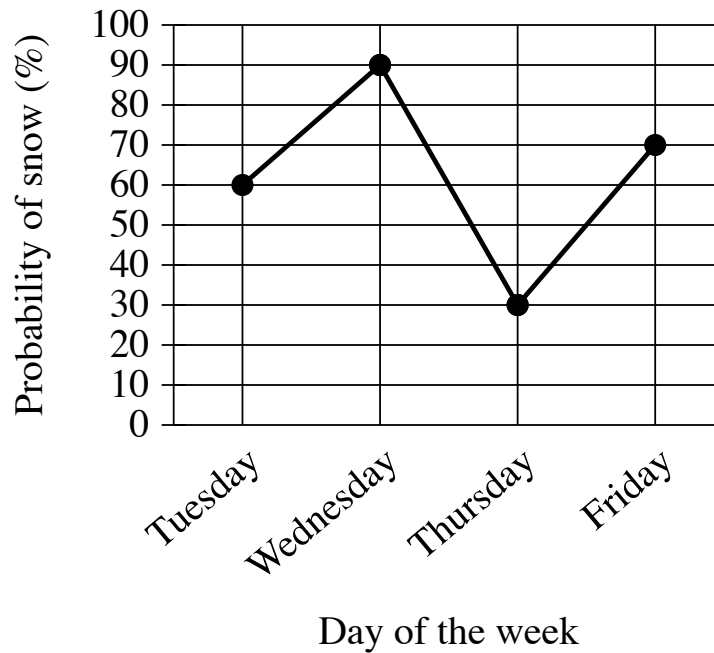
10 questions · ~15 min



- The data for the 5 categories are as follows:
 - 1: More than halfway above 25 students
 - 2: Less than halfway above 30 students
 - 3: More than halfway above 35 students
 - 4: About halfway above 40 students
 - 5: About halfway above 45 students

A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity 3?

- A. 25
- B. 39
- C. 48
- D. 50



- The line graph:
 - Begins at Tuesday, 60%
 - Rises sharply to Wednesday, 90%
 - Falls sharply to Thursday, 30%
 - Rises sharply to Friday, 70%

The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow 30%?

- A. Tuesday
- B. Wednesday
- C. Thursday

D. Friday

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

What length, in centimeters, is equivalent to a length of 51 meters?

1 meter = 100 centimeters

A. 0.051

B. 0.51

C. 5,100

D. 51,000

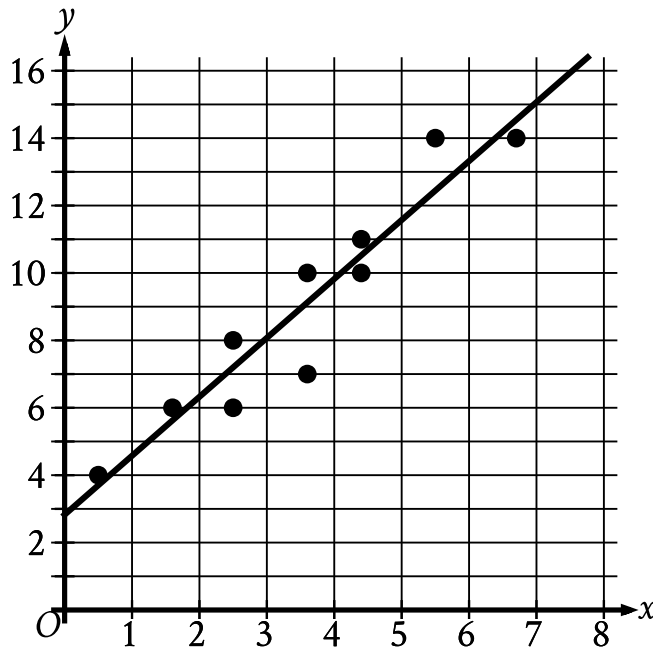
Five Smallest Countries in 2016

Country	Land area (square kilometers)
Monaco	2.0
Nauru	21
San Marino	61
Tuvalu	26
Vatican City	0.44

The table above shows the land area, in square kilometers, of the five smallest countries of the world in 2016. Based on the table, what is the mean land area of the 5 smallest countries in 2016, to the nearest square kilometer?

- A. 20
- B. 22
- C. 61
- D. 110

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown.
 - The line of best fit slants up from left to right.
 - 1 point is touching the line of best fit.
 - 5 points are above the line of best fit.
 - 4 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 3)
 - (3 comma 8)
 - (7 comma 15)

Which of the following equations best represents the line of best fit shown?

- A. $y = 2.8 + 1.7x$
- B. $y = 2.8 - 1.7x$
- C. $y = -2.8 + 1.7x$
- D. $y = -2.8 - 1.7x$

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

- A. \$551.25
- B. \$562.50
- C. \$5,625.00
- D. \$98,000.00

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

2, 2, 2, 3, 4, 4, 11

What is the median of the seven data values shown?

- A. 2
- B. 3
- C. 4
- D. 9

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

An airplane descends from an altitude of 9,500 feet to 5,000 feet at a constant rate of 400 feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A printer produces posters at a constant rate of 42 posters per minute. At what rate, in posters per hour, does the printer produce the posters?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

PROBLEM-SOLVING AND DATA ANALYSIS

Type of store	Average number of employees
Warehouse store	365
Department store	213
Supermarket	130

For a certain region, the table shows the average number of store employees in 2016 by type of store. Based on the table, how much greater was the average number of store employees in warehouse stores than in supermarkets?

- A. 83
- B. 152
- C. 235
- D. 495